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Wise Men on the Track: Unlocking the Multi-layer Secrets of Olympic Medal Prediction

As a global sports event, the Olympic Games not only showcase the outstanding talents of athletes, but also reflect the pattern of international sports competition. Medal allocation, as a key indicator for measuring national sports achievements, has always been a focus of research and analysis. This article aims to establish a **medal quantity prediction model** in order to accurately predict the performance of various countries in the upcoming 2028 Olympic Games.

Firstly, we use the **CNN-LSTM** model to predict the number of events in 2028, and find that the number of events in 2028 is 328. Next, we established an **MLP regression model** based on the **CGO chaotic game optimization algorithm** to predict the number of medals and gold medals for each country in 2028. By calculating the difference in medal scores for each country, we judged their performance compared to 2024. The results are shown in Table 6, and the confidence intervals for the results are shown in Table 7. For example, the predicted total medal range for the United States at a 90% confidence level is **[112,124]**, and the predicted gold medal range is **[52,64]**, showing **better performance compared to 2024**.

Secondly, we predicted the newly awarded countries. Assuming that the number of medals follows a normal distribution, we consider the number of countries that won their first medals in 2024 **as a Poisson distribution** and solve it through **dynamic programming**. The results show that at a **95% confidence level**, the number of countries expected to win at least one medal for the first time in 2028 is **between 14 and 29**. Then, we calculated the correlation between the number of medals won by each country in each event and the number of events, and selected significant events based on **p-values** to evaluate the impact of events on medal numbers. The results are shown in Table 10. For example, **the Aquatics project has a significant impact on 31.6% of countries, while the Squash project only has a significant impact on 0.0% of countries.**

Thirdly, we quantitatively evaluated the impact of 'great coaches' on medals. Taking Chinese volleyball, American volleyball, Romanian gymnastics, and American gymnastics as examples, an **average medal gain model** was constructed, and the results showed that **the medal gain after the coach joined was 1.286 times that before joining**. For France, Australia, and Germany, we found through a comprehensive scoring model that investing in "great coaches" has the highest returns in **women's football** and **women's hockey**, with expected medal gains of **26.58, 26.58, and 21.44**, respectively.

Fourthly, we summarized the key findings of the modeling process: **the Matthew effect in Olympic medal allocation** and **the temporal dynamic characteristics of the "Great Coach" effect**. Based on this, we suggest that the National Olympic Committees consider a coach signing period of **4 to 5 years** and complete the formation of the coaching team **2 to 3 years** before the Olympics.

Finally, we conducted sensitivity analysis on the prediction models for total medals and gold medals, and discussed in depth the advantages and disadvantages of the models.

Keywords: medal prediction; CNN-LSTM; MLP regression model; dynamic programming

Contents

1 Introduction **3**
1.1 Problem Background 3
1.2 Restatement of the Problem 3
1.3 Literature Review 4
1.4 Our Work 5
2 Assumptions and Justifications **5**
3 Notations **6**
4 Prediction of The Number of Events **6**
4.1 Data Pre-processing 6
4.2 Prediction of The Number of Events 7
4.3 Result Analysis 8
5 Prediction of the number of medals and gold medals **9**
5.1 Establishment of Multiple Linear Regression Model 9
5.2 Model Improvement: MLP Regression Model Based on CGO 11
5.3 Performance Evaluation of Various Countries 14
5.4 Confidence Intervals for Medals from Various Countries 15
5.5 Prediction of The Number of Newly Awarded Countries 15
5.6 Exploration of the Relationship between Events and Medals 17
6 'Great Coach' Effect **19**
6.1 Analysis of the 'Great Coach' Effect 19
6.2 Analysis of Potential Countries 20
7 Other Original Insights **22**
7.1 The Matthew Effect in Olympic Medal Allocation 22
7.2 Time Dynamic Characteristics of the "Great Coach" Effect 23
8 Sensitivity Analysis **24**
9 Model Evaluation and Further Discussion **24**
9.1 Strengths 24
9.2 Weaknesses 25
9.3 Further Discussion 25
References **25**

1 Introduction

1.1 Problem Background

As the most influential sports event in the world, the Olympic Games is not only a stage for athletes to showcase their outstanding talents, but also an important embodiment of the sports strength of all countries. The medal table, as the core indicator of a country's sporting achievements, hides a complex intertwining of factors: economic strength, population size, sports culture and event setting, etc., which together shape the pattern of medal distribution. Accurate prediction of medal distribution can not only provide a scientific basis for the decision-making of national Olympic committees, but also open up an important perspective for in-depth research on the laws of sports development.

However, the medal distribution is not set in stone. It changes dynamically with the adjustment of sports policies in various countries, the rise of emerging countries, and the evolution of event settings. Breakthroughs in specific events in emerging countries, traditional powers maintaining their advantages by optimizing their training systems and bringing in excellent coaches, and changes in the event setting have all had a profound impact on the distribution of medals. These dynamics make medal prediction a challenging issue.

Figure 1: Medals for the Paris 2024 Olympics

1.2 Restatement of the Problem

On the basis of fully considering the background of the problem and the established constraints, we need to solve the following problems:

- Develop a predictive model for estimating the number of medals each country has in the Olympic Games, including gold medals and total medals. Provides an estimate of the uncertainty of the predicted results.

- Use the model to predict the medal table for Los Angeles 2028, analysing which countries are likely to progress or regress, and which countries that have not yet won medals are likely to win their first medal in 2028.
- Identify the contribution and potential impact of coach mobility on medal numbers by analyzing historical data. Select three countries to explore what it means to introduce "great coaches" in specific sports.
- Through data analysis, reveal other regular insights into the distribution of medals, discuss how to adjust policies or invest resources to optimize medal performance, and provide decision-making support for NOCs.

1.3 Literature Review

The research on the prediction of the number of Olympic medals at home and abroad mainly includes regression analysis of economic and demographic factors, dynamic system models and machine learning methods. Johnson et al. (2005) [1] proposed a model based on linear regression, taking a country's GDP and population size as the main independent variables, and found that economic strength and population base have a significant positive impact on the number of medals. Glickman and Stern (1998) [2] proposed a model based on Bayesian dynamic systems that treats the change in the number of medals as a dynamic process that evolves over time. In recent years, with the development of machine learning technology, more and more research has begun to employ more sophisticated algorithms to predict the number of Olympic medals. Liu et al. (2016) [3] used a random forest model to predict the medal distribution of the Rio 2016 Olympic Games by combining multidimensional data such as the economy, population, sports culture, and host advantage of the country. The diagram below illustrates the framework of our literature review：

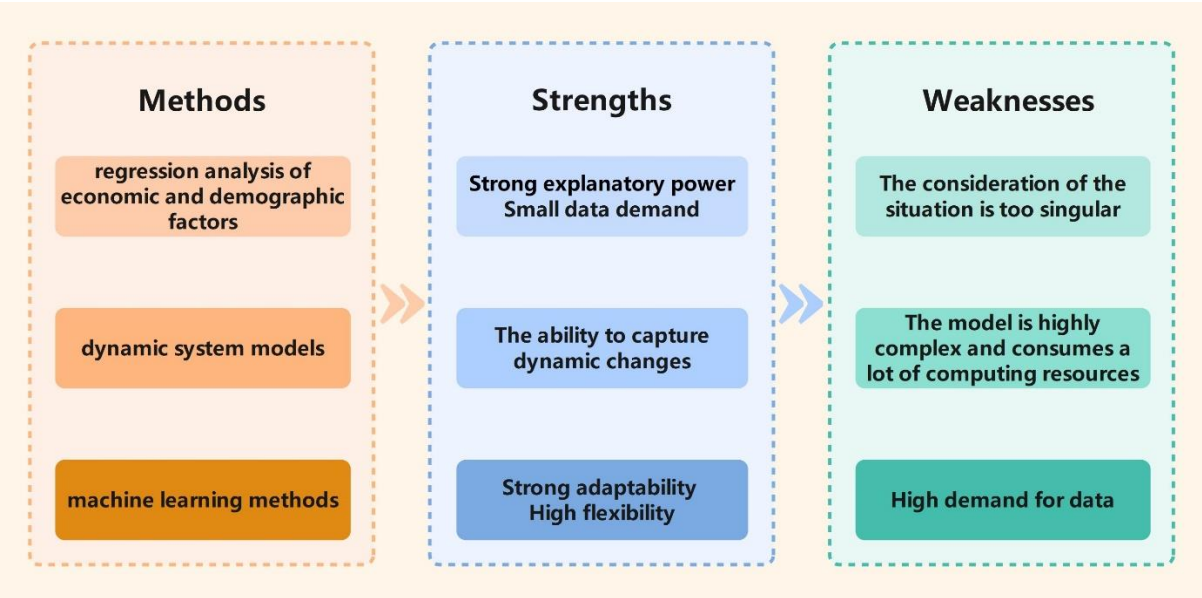


Figure 2: Literature Review Framework

1.4 Our Work

The following figure clearly and thoroughly demonstrates our work:

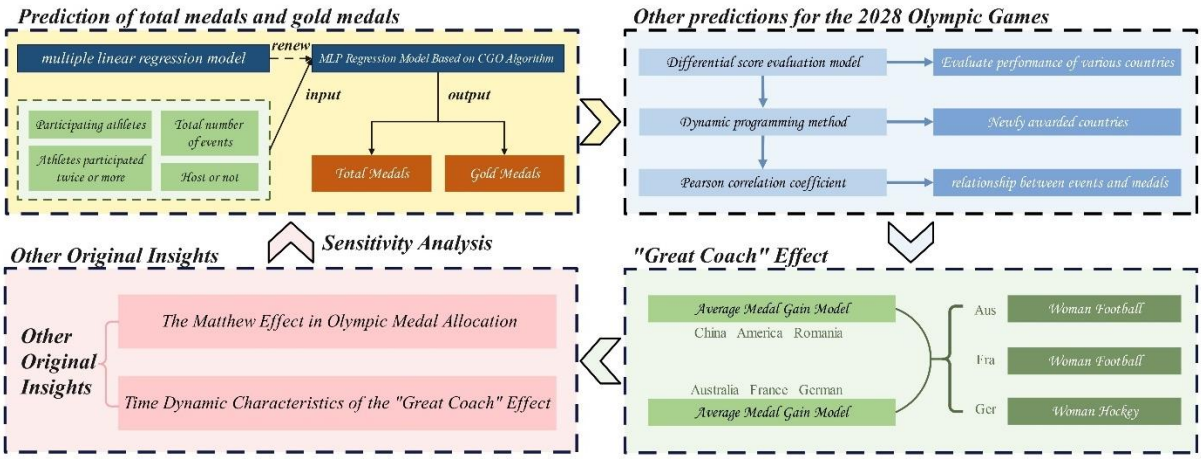


Figure 3: Our work

2 Assumptions and Justifications

1. **The country's sports strength remains relatively stable for a certain period of time.** The sports strength of a country usually has a certain degree of continuity, and historical medal data can serve as an important reference for predicting future medal numbers. This assumption ignores the potential impact of unexpected events such as political turmoil, natural disasters, etc. on sports strength.
2. **The host country has a home advantage in the Olympic Games.** Athletes from the host country are more familiar with the competition environment, can receive strong support from the home audience, and the event schedule may be more conducive to the performance of domestic athletes. In addition, host countries usually increase their investment in sports, improve the quality of athletes' training and competitive ability, in order to achieve better results at the Olympics.
3. **The number of medals is a continuous variable.** We assume that the number of medals awarded by each country is a continuous variable rather than a discrete integer. This assumption simplifies the modeling process, allowing the regression model to handle small fluctuations in medal counts. Although the actual number of medals is an integer, treating it as a continuous variable can more accurately capture the trend of medal numbers, especially when conducting statistical modeling, which is usually treated as a continuous value to improve prediction accuracy.
4. **The probability of winning in different countries is independent.** We assume that the probability of each country winning is independent, which means that the performance of one country will not directly affect the performance of other countries. This assumption simplifies the complexity of the model, allowing us to focus on the individual performance

of each country without considering the complex interdependence between countries, although in some scenarios, the mutual influence between countries cannot be ignored.

3 Notations

The key mathematical notations used in this paper are listed in Table 1. Unclaimed variables are subject to the specific declaration at the symbol occurrence.

Table 1: Notations used in this paper

Symbol	Description
Y	Total number of medals
X_1	Number of athletes
X_2	Number of athletes participating in two or more Olympic Games
X_3	The host or not
X_4	Total events
$\beta_0, \beta_1, \ldots, \beta_4$	Multiple linear regression coefficients
ϵ	Random error term
$Diff$	Differential score
r	Pearson correlation coefficient

4 Prediction of The Number of Events

4.1 Data Pre-processing

Since we are only allowed to use datasets provided by COMAP, which were recorded by the International Olympic Committee (IOC) during a given Olympic Games, there is a possibility that the data will be anomalous. Therefore, this data should be preprocessed before the model is built. The steps of data preprocessing are as follows:

- **Missing value handling:** Fill in missing values in the dataset and replace them with 0 uniformly. In addition, for years missing due to the cancellation of Olympic competitions (such as 1916, 1940, and 1944), their respective data will also be filled in with 0. Missing values and cancelled Olympic years both indicate no medal production during these time periods. Filling in with 0 can maintain data integrity and consistency, while avoiding analysis bias caused by missing values.
- **Elimination of informal Olympic data:** Remove relevant data from the 1906 informal Olympic Games. The 1906 competition was not officially recognized as an Olympic Games by the International Olympic Committee, and its data was inconsistent with the statistical system of the official Olympic Games, which may interfere with subsequent analysis.
- **Standardization of country names:** Unify the handling of inconsistencies in the names of some countries in the dataset. For example, unifying "Federal Germany" into "Germany". This standardization process can ensure data consistency, avoid statistical errors caused by differences in country names, and thus improve the accuracy of data analysis.
- **Medal quantification processing:** Gold medals are quantified as 5 points, silver med-

als and bronze medals are quantified as 1 point. This quantitative approach can incorporate the value differences of different types of medals into the analysis system, while providing more intuitive numerical basis for subsequent statistical analysis and model construction.

4.2 Prediction of The Number of Events

In Olympic medal prediction, the number of events is a key factor. The change in the number of events directly affects the distribution of medals among countries, as more events mean more medal opportunities. Therefore, accurately predicting the number of events for the 2028 Los Angeles Olympics is crucial for the construction of medal prediction models. However, in our dataset, the number of events for 2028 was not provided, which posed a challenge for subsequent work.

However, we found through online searches that the number of sports for the 2028 Los Angeles Olympics is 35, but the specific number of events has not been clearly announced. To fill this data gap, we adopted the CNN-LSTM model, which is a deep learning model that combines convolutional neural networks (CNN) and long short-term memory networks (LSTM). It can effectively handle complex patterns and long-term dependencies in time series data, providing a reasonable prediction for the number of events for the 2028 Olympic Games.

The CNN-LSTM model combines the feature extraction capability of CNN with the time series modeling capability of LSTM. The CNN part is used to extract local features from time series data, while the LSTM part is used to capture long-term dependencies in time series. This combination model is particularly suitable for handling data with temporal and spatial dependencies, such as predicting the number of Olympic events. The principle of the CNN-LSTM model is shown in the following figure:

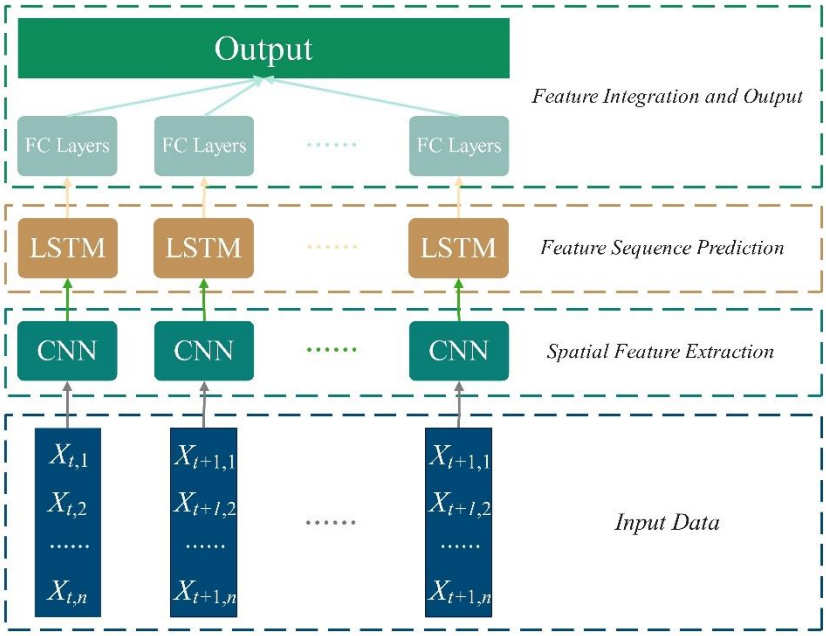


Figure 4: Principle of CNN-LSTM model

Step 1: Build a CNN-LSTM model. This model combines the feature extraction capability of Convolutional Neural Networks (CNN) with the time series modeling capability of Long-

Short-Term Memory Networks (LSTM). Choose appropriate hyperparameters, such as convolution kernel size, number of LSTM units, etc., to ensure that the model can effectively learn complex patterns in the data.

Step 2: Train the CNN-LSTM model using historical data. We adopted a cross validation approach, dividing the data into training, validation, and test sets, adjusting model parameters, and optimizing model performance. Record evaluation metrics including mean absolute error (MAE), root mean square error (RMSE), and coefficient of determination (R^2).

Step 3: After model training and validation, the trained CNN-LSTM model was used to predict the number of events for the 2028 Olympic Games. To provide confidence intervals for the predicted results, we evaluated the uncertainty of the predictions. This not only provides a predicted number of events in 2028, but also offers a reliable evaluation of the predicted results, providing important reference for subsequent medal prediction models.

Through the above method, we predict that the total number of events for the 2028 Los Angeles Olympics will be 311. The following table lists the detailed information of the predicted results:

Table 2: Detailed information of the predicted results

Parameter	Training Set	Validation Set	Test Set
MAE	10.1639	17.0332	7.6775
MAPE	0.070685	0.056192	0.022813
MSE	191.0629	295.2063	65.6791
RMSE	13.8226	17.1816	8.1043
R^2	0.95324	0.40268	0.63556

It is not difficult to see from the above table that the R^2 values of the training set, validation set, and test set are all between 0 and 1, indicating that the model has a good fitting effect and accurate prediction results. In order to more intuitively demonstrate the degree of fit between predicted data and real data, we have drawn images of predicted data and real data for the training set, validation set, and test set, as shown in the following figure:

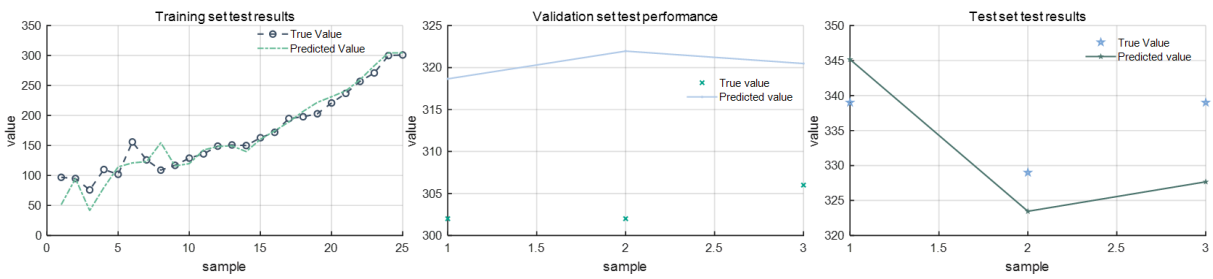


Figure 5: The real and predicted values of the events

The figure above clearly reflects that the predicted values of the number of events in the training set, validation set, and testing set can all fit the actual data well, further proving the rationality of the training results. **Ultimately, we determine that the number of events for 2028 will be 328**.

4.3 Result Analysis

Although we expect the number of events for the 2028 Olympics to increase due to an

increase in sports, the actual forecast is 328, lower than the data for 2024. This result may be influenced by multiple factors and has a certain degree of rationality.

Firstly, historical data indicates that changes in the number of events are not always completely consistent with changes in the number of sports. For example, the number of sports at the 1964 Tokyo Olympics decreased from 19 in 1965 to 18, but the number of events in 1964 was 163 and the number of events in 1965 increased to 172. This example clearly illustrates that even if the number of sports events decreases, the number of events may still increase, and vice versa. This phenomenon indicates that changes in the number of events are influenced by a combination of multiple factors

In recent years, the International Olympic Committee has been committed to promoting the sustainable development and streamlined management of the Olympic Games. In this context, we speculate that the number of events may be optimized and adjusted to ensure the quality and sustainability of the events. For example, it may merge some similar projects or make adjustments to some projects with lower participation. This adjustment helps to improve the overall quality and attractiveness of the event, which is in line with the development direction of the Olympic Games.

In addition, the prediction model itself also has a certain degree of uncertainty. Although we have adopted the advanced CNN-LSTM model, the prediction results of the model are still affected by the limitations of historical data and future uncertainty factors. Therefore, it is normal for there to be some deviation between the predicted results and the actual number of events, which also reflects the complexity and challenge of prediction work.

In summary, although the predicted number of events for the 2028 Olympics is lower than the data for 2024, this result has a certain degree of rationality in the context of historical data fluctuations and the International Olympic Committee's adjustment strategies. This also reminds us that when predicting the number of events, we need to fully consider the comprehensive impact of multiple factors and recognize the uncertainty of the prediction results.

5 Prediction of Total Medals and Gold Medals

5.1 Establishment of Multiple Linear Regression Model

By analyzing the data tables provided by COMAP, we have identified the key factors that affect the number of medals:

1. **The number of participating athletes:** The number of athletes who participated in the Olympic Games directly affects the potential chances of winning medals. More participating athletes mean more medal competition points, so it has a significant positive impact on the number of medals.
2. **Athletes who have participated in two or more consecutive Olympic Games:** Athletes who have participated in two or more consecutive Olympic Games often have richer competition experience and more mature competitive state. They are more competitive in the competition, and their medal data is more referenceable. Therefore, this variable has important reference value for predicting the number of medals.
3. **Host advantage:** The host country usually has a home advantage in the Olympic Games,

including familiarity with the competition environment, audience support, and understanding of the event schedule, all of which contribute to increasing the number of medals won by the host country. Therefore, it is reasonable to set the host variable as a dummy variable of 0 or 1.

4. **The total number of events in the Olympic Games:** This variable determines the basis for medal allocation, and changes in the event settings of different Olympic Games will directly affect the medal winning opportunities of each country. Therefore, using the total number of events as an input variable can better reflect the trend of medal quantity changes.

Here, we take the prediction model of the total number of medals as an example to establish the model. Perform linear regression analysis using the above four variables as input features and the total number of medals as output features.

Establish a multiple linear regression model:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \beta_4 X_4 + \epsilon \tag{1}$$

In this model, Y is the dependent variable (total number of medals), X_1, X_2, X_3, X_4 are independent variables (number of athletes, number of athletes participating in two or more consecutive events, whether they are the host country, total number of events), $\beta_0, \beta_1, \beta_2, \beta_3, \beta_4$ are regression coefficients, and ϵ is the random error term.

Using MATLAB to solve the model, the values of $\beta_0, \beta_1, \beta_2, \beta_3$ and β_4 are obtained as 0.0000, 0.9628, -0.1527, 0.0240, and -0.0308, respectively:

$$Y = 0.9628X_1 - 0.1527X_2 + 0.0240X_3 - 0.0308X_4 \tag{2}$$

In this model, the coefficient of the number of athletes is 0.9628, which is the highest among all independent variables. This indicates that the number of athletes has the most significant impact on the number of medals, highlighting the importance of the number of participants in medal competition. In contrast, the coefficient for the number of athletes participating in two or more consecutive events is -0.1527, which is negative, indicating that this factor has a relatively small impact on the number of medals and may have a certain inhibitory effect. The coefficient of the host factor is 0.0240, indicating that the host advantage has a certain promoting effect on the number of medals, but the degree of influence is relatively limited. Finally, the coefficient for the total number of events is -0.0308, which is also negative, indicating that an increase in the number of events has a relatively small impact on the number of medals, and may even have a slight negative effect. Overall, the number of athletes is a key factor affecting the number of medals, while the influence of other factors is relatively small.

The fitting parameters of the multiple linear regression model are shown in the following table:

Table 3: Fitting parameters of the multiple linear regression model

Parameter	Training Set	Validation Set	Test Set
MAE	4.2591	3.7335	4.1885
MAPE	Inf	Inf	Inf
MSE	79.9145	43.6676	84.7521
RMSE	8.9395	6.6081	9.2061
R^2	0.69806	0.62434	0.59871

In addition, the linear correlation coefficients of the model on the training set, testing set,

and validation set are 0.69806, 0.75295, and 0.60306, respectively. These results indicate that the fitting effect of the multiple linear regression model is not ideal, especially with a linear correlation coefficient of only 0.60306 on the validation set, demonstrating significant limitations of the model in explaining the variation of the dependent variable (medal count).

The multiple linear regression model assumes a linear relationship between the dependent variable and the independent variable, but in actual data, there may be non-linear relationships or complex interactions between the dependent variable and the independent variable, which cannot be effectively captured by linear models. For example, the relationship between the number of athletes and the number of medals may not be a simple linear relationship, and there may be threshold effects or other nonlinear features.

Given the limitations of the multiple linear regression model, we have instead introduced a multi-layer perceptron (MLP) regression model to more accurately predict the number of medals.

5.2 Model Improvement: MLP Regression Model Based on CGO

Compared to multiple linear regression models, MLP (Multilayer Perceptron) regression models have significant advantages. The MLP regression model is a neural network-based non-linear model that can capture complex nonlinear relationships and interactions in data, while the multiple linear regression model assumes a linear relationship between the dependent variable and the independent variable, which cannot effectively handle nonlinear relationships.

In our case, the MLP regression model can better adapt to the complexity of predicting medal numbers, as it can automatically learn hidden patterns and features in the data without the need for human assumptions about the relationships between variables. In addition, the MLP regression model has stronger generalization ability and can more effectively handle new and unseen data, thereby improving the accuracy and reliability of predictions.

MLP (Multilayer Perceptron) is a typical artificial neural network model widely used in regression and classification tasks. MLP consists of multiple neurons, typically including an input layer, one or more hidden layers, and an output layer. It is a type of feedforward neural network, so information is passed from the input layer to the output layer without feedback connections.

Assuming there is an MLP with L layers, where the output of each layer passes through an activation function (except for the output layer), for the lth layer ($l = 1,2,\ldots,L-1$), the calculation formula for the hidden layer is:

$$h^{(l)} = f^{(l)}(W^{(l)}h^{(l-1)} + b^{(l)}) \tag{3}$$

In the above expression, $h^{(l-1)}$ is the output of the previous layer (which is the input data x for the input layer), $W^{(l)}$ is the weight matrix of the lth layer, and $b^{(l)}$ is the bias vector of the lth layer, $F^{(l)}$ is the activation function of this layer.

For the output layer (Lth layer), the calculation formula is:

$$\hat{y} = W^{(L)}h^{(L-1)} + b^{(L)} \tag{4}$$

Here, $\hat{y}$ is the final predicted value. The MLP network updates weights and biases through backpropagation, with the goal of minimizing the loss function. Use gradient descent to update parameters:

$$\begin{cases} W^{(L)} = W^{(L)} - \eta \dfrac{\partial \varphi}{\partial W^{(L)}} \\ b^{(L)} = b^{(L)} - \eta \dfrac{\partial \varphi}{\partial b^{(L)}} \end{cases} \tag{5}$$

Among them, η is the learning rate, $\frac{\partial \varphi}{\partial W^{(L)}}$ and $\frac{\partial \varphi}{\partial b^{(L)}}$ are the gradients of the loss function with respect to weights and biases. After forward propagation through a multi-layer perceptron, the calculation formula for the output predicted value $\hat{y}$ is:

$$\hat{y} = W^{(L)} f^{(L-1)}\left(W^{(L-1)} f^{(L-2)}\left(\ldots f^{(1)}\left(W^{(1)} x + b^{(1)}\right)\ldots\right) + b^{(L-1)}\right) + b^{(L)} \tag{6}$$

By continuously adjusting weights and biases, MLP can optimize prediction results.

However, conventional MLP regression models have some shortcomings. Firstly, gradient descent or other traditional optimization algorithms are prone to getting stuck in local optima, especially when dealing with complex nonlinear problems, making it difficult to find global optima. Secondly, overfitting is common, especially in high-dimensional data or when there is a lot of noise, the model overfits the training data, resulting in poor performance on the validation and test sets. In addition, traditional optimization algorithms are sensitive to initial parameters, and improper initial parameters may lead to low optimization efficiency and even converge to poor solutions.

To address these issues, we introduced the CGO chaotic game optimization algorithm to optimize the parameters of the MLP regression model. The CGO algorithm avoids local optima through global search, utilizes the nonlinear characteristics of chaotic sequences and the competitiveness of game strategies to explore a wide solution space and find the global optimal solution. This enables the MLP regression model to more accurately fit complex nonlinear data, improve prediction accuracy and generalization ability, and ensure that the model can provide stable and reliable prediction results on training, validation, and testing sets.

In addition, we used visualization methods to more intuitively demonstrate the principle of the MLP regression model based on CGO algorithm:

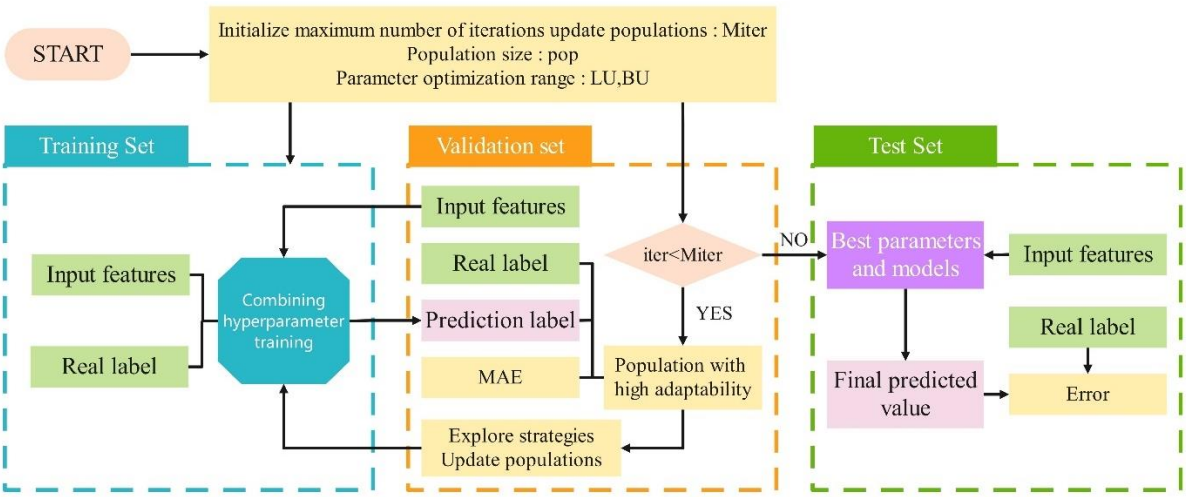


Figure 6: The principle of the MLP based on CGO algorithm

Through the above method, we have predicted the total number of medals for each country

at the 2028 Los Angeles Olympics. The following table lists the various parameters of the predicted results:

Table 4: Various parameters of the predicted results

Parameter	Training Set	Validation Set	Test Set
MAE	3.0006	2.312	3.3381
MAPE	Inf	Inf	Inf
MSE	63.9627	26.8764	74.2631
RMSE	7.9977	5.1842	8.6176
R^2	0.7454	0.84331	0.71966

It can be clearly seen from the above table that compared to the multiple linear regression model, the improved model has significantly improved the R^2 value, indicating a significant improvement in the fitting effect of the model. In order to demonstrate the fitting effect of the improved model more intuitively, we plotted the real and predicted values of the training set, validation set, and test set separately, as shown in the following figure:

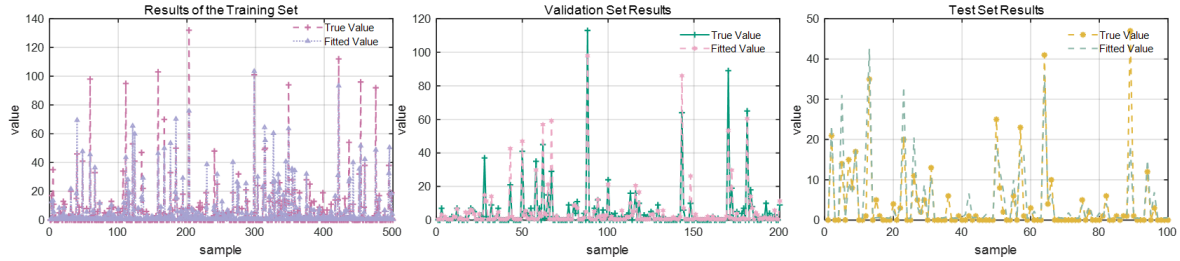


Figure 7: Real and predicted values of training set, validation set, and test set

The results showed that the fitting effect of the model was good, and the deviation between the actual and predicted values of the total number of medals for each country was low. Similarly, for predicting the number of gold medals won by each country in 2028, the parameters are as follows:

Table 5: Various parameters of the predicted results

Parameter	Training Set	Validation Set	Test Set
MAE	1.2772	1.2293	1.1426
MAPE	Inf	Inf	Inf
MSE	12.7998	12.0492	7.1854
RMSE	3.5777	3.4712	2.6806
R^2	0.65188	0.71876	0.51692

The real and predicted value images of its training set, validation set, and test set are shown in the following figure:

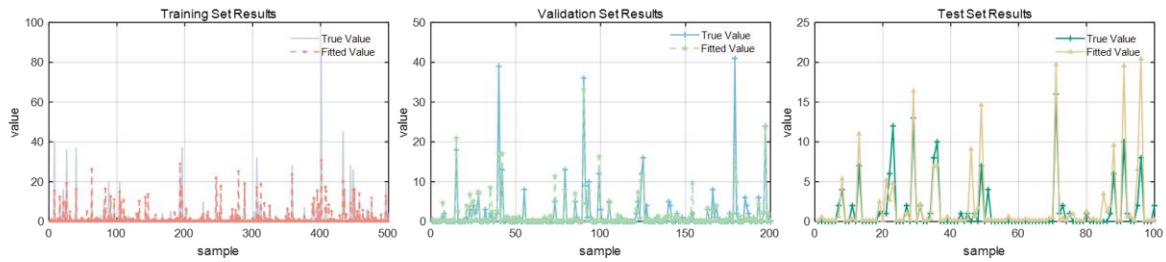


Figure 8: Real and predicted values of training set, validation set, and test set

5.3 Performance Evaluation of Various Countries

In addition, to evaluate which countries are most likely to improve their award situation and which countries may perform worse than 2024, we have defined a national performance evaluation indicator. Considering that the award situation of each country is mainly determined by the number of gold medals and the total number of medals, we set the difference in the number of gold medals as 5 points, and the difference in the number of other medals as 1 point, thus obtaining the following formula for calculating the difference points:

$$Diff = 5 \cdot GoldDiff + 1 \cdot (TotalDiff - GoldDiff) \tag{7}$$

Among them, $GoldDiff$ represents the difference between the number of gold medals in 2028 and 2024, $TotalDiff$ represents the difference between the total number of medals in 2028 and 2024, and $Diff$ represents the difference score. When $Diff$>0, the country performs better than in 2024; When $Diff$=0, the country performs the same as in 2024; When $Diff$<0, the country's performance is worse than in 2024.

We have predicted the number of Olympic medals and gold medals for each country in 2028. Due to the inability to display all countries' awards, we only selected the top eight countries with the highest number of medals as the medal table for display. The results are shown in the following table:

Table 6: Various parameters of the predicted results

NOC	USA	GER	AUS	CHN	JPN	ITA	CAN	GBR
Total medals	118	94	92	65	55	49	47	47
Gold medals	58	25	25	22	19	17	16	16
Diff	64	113	67	-98	6	29	48	-10
Performance	↑	↑	↑	↓	↑	↑	↑	↓

From the table above, it can be seen that among the 8 countries presented in the results, the predicted performance of the 2028 Olympics of the United States, Germany, Australia, Japan, Italy, and Canada will be better than in 2024, while the predicted performance of China and the United Kingdom in 2028 will all decline compared to 2024.

Furthermore, we plotted the distribution of medals and gold medals among countries from 1984 to 2024, as shown in the following figure:

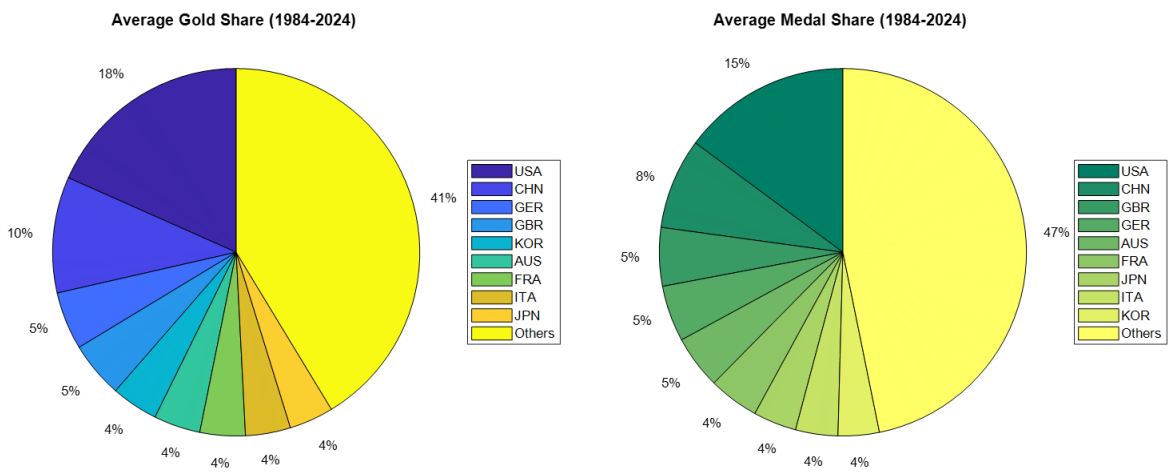
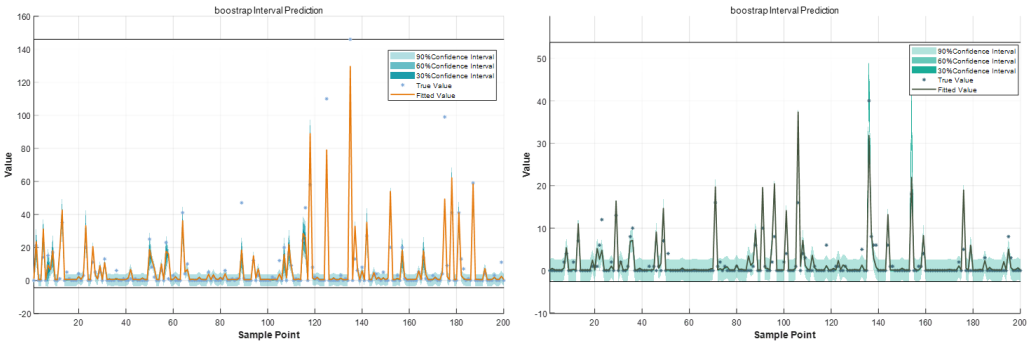


Figure 9: Distribution of medals and gold medals among countries

It can be clearly seen from the chart that the 8 countries with the highest number of medals predicted by us have high consistency with historical data, which further verifies the accuracy of our model and the rationality of our predictions.

5.4 Confidence Intervals for Medals from Various Countries

In our model solving process, not only can we predict the number of medals and gold medals in each country, but we can also simultaneously solve the confidence intervals of these predicted values. Specifically, we calculated confidence intervals at 90%, 60%, and 30% confidence levels to assess the uncertainty of the prediction results. We have plotted the bootstrap interval prediction graph of the number of medals and gold medals for each country, as shown in the following figure:



(a) Total medals (b) Gold medals

Figure 10: Bootstrap interval prediction

Both of these images demonstrate good fitting results. Although there are some wide confidence intervals in the left figure, especially in the high peak area, the fitted curve is still close to the true value, indicating that the model can capture most of the trends in the data. The figure on the right shows a smoother fitting curve with a relatively narrow confidence interval and lower prediction uncertainty, further demonstrating the accuracy and stability of the model with high confidence.

Similarly, we present the calculated confidence intervals for total medals and gold medals of the 8 countries mentioned above, as shown in the table below:

Table 7: Calculated confidence intervals for total medals and gold medals

NOC	Confidence interval for total medals			Confidence interval for gold medals		
	90% interval	60% interval	30% interval	90% interval	60% interval	30% interval
USA	[112, 124]	[112, 124]	[112, 124]	[52, 64]	[52, 64]	[52, 64]
GER	[88, 100]	[88, 100]	[88, 100]	[22, 30]	[23, 29]	[24, 27]
AUS	[86, 98]	[86, 98]	[86, 98]	[21, 30]	[22, 29]	[23, 27]
CHN	[49, 74]	[52, 72]	[59, 67]	[19, 27]	[19, 27]	[21, 24]
JPN	[39, 64]	[42, 62]	[49, 57]	[15, 24]	[16, 23]	[18, 21]
ITA	[32, 57]	[36, 55]	[43, 51]	[13, 22]	[14, 21]	[16, 19]
CAN	[30, 55]	[34, 53]	[41, 49]	[12, 21]	[13, 20]	[15, 18]
GBR	[31, 55]	[34, 53]	[41, 49]	[13, 21]	[14, 20]	[15, 18]

5.5 Prediction of The Number of Newly Awarded Countries

To predict the number of medals won by countries that have not won medals for the first

time in the next Olympic Games, we first assume that the number of medals is continuous and follows a normal distribution. Based on this assumption, we can calculate the probability that each country will win at least one medal in 2028. This probability can be calculated using the cumulative distribution function (CDF) of a normal distribution, with the specific formula being:

$$P(X>1)=1-\Phi\left(\frac{1-\mu}{\sigma}\right) \tag{8}$$

Among them, Φ represents the cumulative integral function of the standard normal distribution, and μ and σ are the mean and standard deviation of the number of medals, respectively. Next, we will consider the Poisson binomial distribution problem as' how many countries that have not won previously will win their first gold medal in 2028 '. The Poisson binomial distribution describes the distribution of success rates in a series of independent Bernoulli experiments, where the success probability of each experiment can vary. In our case, each non winning country winning at least one medal in 2028 can be considered as a Bernoulli experiment with a success probability of $P(X>1)$. We use dynamic programming to accurately calculate the probability $P(X=k)$ of k countries winning their first prize.

By solving the above model, we obtained the probability corresponding to each possible k and the expected value $E(X)$. The results show that it is expected that 21.46 countries will receive their first award in 2028 (expected value). At a 95% confidence interval, the number of first-time winners is approximately between [14, 29]. Below, we have predicted some newly awarded countries, as shown in the table below:

In order to accurately calculate the probability of Poisson binomial distribution, we use dynamic programming method. The method of dynamic programming can effectively calculate the probability of Poisson binomial distribution, avoiding the complexity of direct calculation. The specific steps are as follows:

Step 1 Initialization: Define a two-dimensional array $dp[i][j]$, where $dp[i][j]$ represents the probability of j out of the top i countries winning for the first time. Initialize $dp[0][0]=1$, indicating that the probability of no country winning the award is 1.

Step 2 State transition: For each country i, update the value of $dp[i][j]$ based on its winning probability p_i. The specific formula is:

$$dp[i][j]=dp[i-1][j-1]\cdot p_i+dp[i-1][j]\cdot(1-p_i) \tag{9}$$

Among them, $dp[i-1][j-1]\cdot p_i$ represents the ith country that won the award, while $dp[i-1][j]\cdot(1-p_i)$ represents the ith country that did not win the award.

Step 3 Calculate the expected value: The expected value $E(X)$ can be calculated by multiplying all j by $P(X=j)$ and then summing them up, that is:

$$E(X)=\sum_{j=0}^{n} j\cdot P(X=j) \tag{10}$$

Among them, n is the total number of countries that have not won awards.

By solving the model above, we obtained the probability corresponding to each possible k and the expected value $E(X)$. The results show that it is expected that 21.46 countries will receive their first award in 2028 (expected value). At a 95% confidence interval, the number of first-time winners is approximately between [14, 29]. Below, we have predicted some newly

awarded countries, as shown in the table below:

Table 8: Newly awarded countries

NOC	Predicted medals	90% interval	60% interval	30% interval	Probability
ANG	1.784	[-2.9,5.1]	[1.0,2.3]	[1.3,1.4]	0.760
HON	1.099	[-3.6,4.4]	[0.4,1.6]	[0.6,0.7]	0.535
SSD	0.853	[-3.8,4.2]	[0.1,1.4]	[0.3,0.4]	0.447
MLI	0.696	[-4.0,4.0]	[0.0,1.2]	[0.2,0.3]	0.392
SAM	0.677	[-4.0,4.0]	[0.0,1.2]	[0.1,0.3]	0.386

These results indicate that although 77 countries have not yet won an Olympic medal, it is expected that around 21 countries will make their first breakthrough and win at least one medal in 2028. This prediction is based on the probability of winning in each country, taking into account various factors such as the country's level of sports development and the number of participating events. The calculation of the 95% confidence interval further illustrates the uncertainty of the prediction results and provides a more comprehensive evaluation perspective.

5.6 Exploration of the Relationship between Events and Medals

To explore the relationship between the number of events and the number of medals won by a country, we first need to establish a mathematical model to describe this relationship. We assume that there is a linear relationship between the number of medals won by each country in each sport and the number of events in that sport, which means that Pearson correlation coefficient can be used to measure the strength and direction of this relationship.

The calculation formula for Pearson correlation coefficient r is as follows:

$$r=\frac{\sum_{i=1}^{n}\left(X_i-\bar{X}\right)\left(Y_i-\bar{Y}\right)}{\sqrt{\sum_{i=1}^{n}\left(X_i-\bar{X}\right)^2}\sqrt{\sum_{i=1}^{n}\left(Y_i-\bar{Y}\right)^2}} \tag{11}$$

Among them, X_i and Y_i are the number of events and medals in the ith Olympic Games, $\bar{X}$ and $\bar{Y}$ are their averages, and n is the number of Olympic Games.

By calculating the Pearson correlation coefficient r and corresponding p-value between the number of medals won by each country in each sport and the number of events in that sport, we can evaluate the degree of impact of each sport on the number of medals won by the country. Items with a p-value less than 0.05 are considered significantly correlated.

Using MATLAB software, calculate the Pearson correlation coefficient r and corresponding p-value between the number of medals won by each country in each sport and the number of events in that sport. Remove data with p-values greater than 0.05 and retain significantly correlated items. Given the large number of countries and projects, it is difficult to list them all. Therefore, this article only selects Belgium and China as examples to present some of the results:

Table 9: Pearson correlation coefficient

Belgium			China		
event	r	p-value	event	r	p-value
Modern Pentathlon	-0.444	0.01782	Weightlifting	0.856	4.742e-05
Sailing	0.459	0.01406	Gymnastics	0.862	3.67e-05
Rugby	0.471	0.01137	Table Tennis	0.865	3.073e-05
Wrestling	-0.501	0.00667	Archery	0.892	8.078e-06
Canoeing	-0.507	0.00592	Sailing	0.898	5.386e-06
Gymnastics	-0.518	0.00472	Fencing	0.900	4.846e-06
Basketball	-0.538	0.00312	Judo	0.925	8.073e-07
Polo	0.644	0.00022	Athletics	0.929	5.418e-07
Tug Of War	0.645	0.00021	Cycling	0.932	4.274e-07
Archery	0.652	0.00016	Aquatics	0.938	2.287e-07

In the table above, we selected the top ten sets of data with the highest linear correlation between the two countries. It can be seen that the Pearson correlation coefficient r values of China in multiple events are very close to 1, and the corresponding p-values are extremely small, indicating a strong positive correlation between these events and China's medal count, reflecting China's strong competitive strength and stable performance in these events. In contrast, Belgium's correlation indicators are relatively scattered, with some projects even showing negative correlations. This difference may stem from differences in the allocation of sports resources, training systems, and athlete selection mechanisms between the two countries. Through this comparative analysis, we can gain a more comprehensive understanding of the performance differences of different countries in the Olympic Games, and provide reference for countries to formulate sports development strategies.

Finally, we selected events that have a significant impact on the medal count of most countries ($\geqslant$ 10%), as well as events that have little impact on the medal count of most countries (only<10% of countries have a significant impact). Due to space limitations, only 5 items are listed in the following table:

Table 10: Significant events

Influential	Proportion	Uninfluential	Proportion
Aquatics	31.6%	Squash	0.0%
Athletics	30.9%	Flag Football	0.0%
Boxing	31.6%	Rackets	0.0%
Triathlon	32.2%	Lacrosse	1.3%
Fencing	31.6%	Roque	1.3%

It can be clearly seen from the above table that the events that have a significant impact on the number of national medals are mostly popular, which usually have wide participation and high attention, attracting a large number of countries to invest resources in training and competition. On the contrary, events with less impact are mostly unpopular, with relatively few participating countries and limited attention and resource investment. This difference reflects the uneven popularity and resource allocation of different events worldwide, and also reminds

us to fully consider the popularity and participation of sports when analyzing the number of national medals.

6 'Great Coach' Effect

6.1 Analysis of the 'Great Coach' Effect

Coaches play a crucial role in sports competition, especially those known as' great coaches' whose experience and expertise can significantly enhance athletes' performance. For example, Lang Ping led the US and Chinese women's volleyball teams to championships, while Béla Károlyi achieved great success with the Romanian and US women's gymnastics teams. These examples indicate that coach mobility and influence may have a significant impact on medal count. Therefore, analyzing the "great coach" effect and its contribution to the number of medals is of great significance.

In order to analyze the 'Great Coach' effect, we collected medal data before and after Lang Ping and Béla Károlyi joined different countries. Due to the small amount of data, regression or other complex methods (with many features) may lead to significant errors. Therefore, we chose to make some rough estimates to avoid overfitting and model instability issues. We only consider the medal scores during the coach's tenure and the year before the coach's arrival, without taking into account the impact after the coach's departure.

Define an indicator event average medal gain α and construct an average medal gain model as follows:

$$\alpha = \frac{S_2/N_2 + S_1/N_1 - S_0/N_0}{2} \tag{12}$$

Among them, S_2 represents the sum of scores during the coach's tenure, and N_2 represents the number of matches during the coach's tenure; S_1 represents the score of the coach's first year of service, and N_1 represents the number of matches played during the first year of service; S_0 represents the score of the coach in the year before taking office, and N_0 represents the number of matches in the year before taking office.

It should be noted that due to the limited amount of data here, we set the score for the silver medal to 2 points, while keeping the scores for the gold and bronze medals unchanged. Therefore, the calculation method for the score during the tenure period in the above equation is as follows:

$$S_i = 5 \times GoldMedals + 2 \times SilverMedals + 1 \times BronzeMedals \tag{13}$$

We plotted the medal scores of Chinese volleyball, USA volleyball, Romanian gymnastics, and USA gymnastics coaches before and after their arrival, as shown in the following figure:

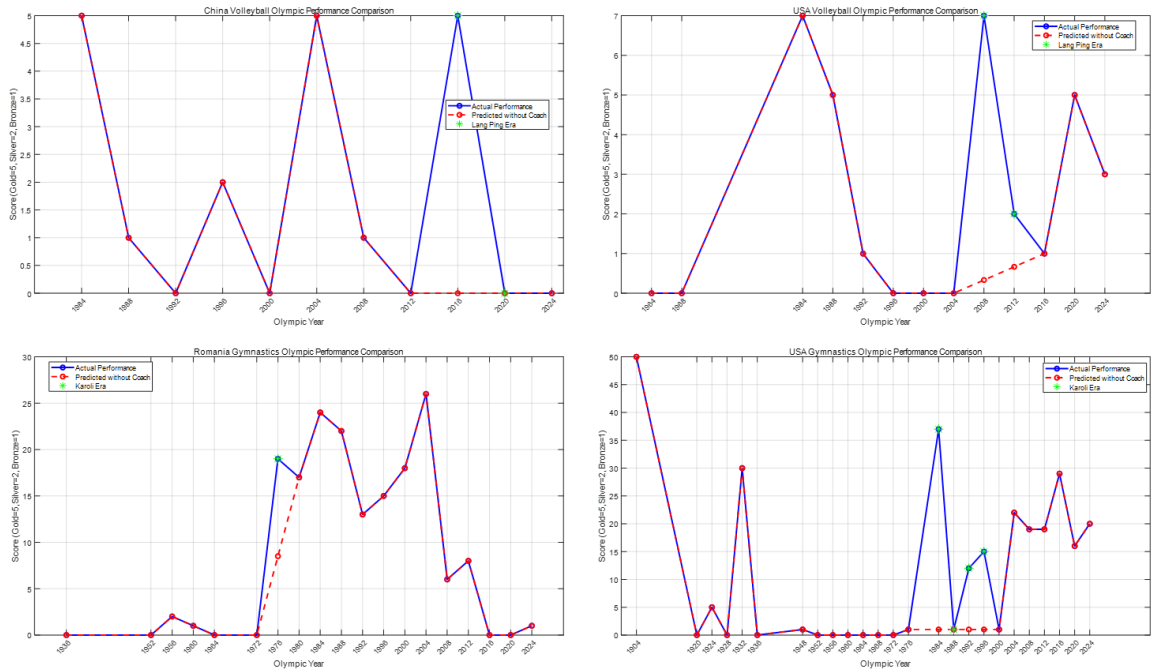


Figure 11: Medal scores

Calculate the average medal gain α for Chinese volleyball, USA volleyball, Romanian gymnastics, and USA gymnastics, as well as the Pearson correlation coefficient between the number of medals and the exercise before the coach's arrival. The calculation results are shown in the table below：

Table 11: Calculation results

NOC sport	α	r	p-value
CHN volleyball	1.000	0.9129	3.399e-5
USA volleyball	1.375	0.1544	0.4714
ROU gymnastics	1.071	0.7138	0.0136
USA gymnastics	1.698	0.2148	0.3921

The above table shows that before and after the coach's arrival, the medal scores of USA volleyball, Romanian gymnastics, and USA gymnastics all increased, while Chinese volleyball remained the same as before. From the perspective of Pearson correlation coefficient analysis, the increase in the number of events is strongly correlated with the increase in Chinese volleyball medal scores, but has a relatively small impact on the sports of the other two countries. We infer that this is due to volleyball being a traditional sport in China, and it can also be seen that before the coach arrived, USA volleyball, Romanian gymnastics, and USA gymnastics were all promising sports.

6.2 Analysis of Potential Countries

In order to screen for countries with the potential to achieve better results at the 2028 Olympics, we first established screening criteria: we focused on countries with a certain level of participation scale (more than 200 athletes) and a certain overall medal rate (less than 25%, greater than 15%). These standards aim to identify countries that have both fundamental strength and significant room for improvement. Based on these criteria, we screened all participating countries and ranked them in descending order of the number of athletes, ultimately

selecting the top three countries:

Table 12: Top three countries

NOC sport	2024 total athletes	2024 total medals	Medal rates
FRA	801	187	0.233458
AUS	649	121	0.186441
GER	647	108	0.166924

Ultimately, we chose France, Australia, and Germany for further research. In order to evaluate the medal winning potential of various countries in specific sports events, we constructed a comprehensive scoring model. This model comprehensively considers four key factors: the number of participants, medal gain, event capacity, and investment effectiveness coefficient: the number of participants reflects the country's participation and resource investment; Medal gain indicates a positive correlation between the number of medals and their potential; The capacity of the event is calculated by dividing the total number of medals in 2024 by 3, which reflects the distribution of medals in the event. However, this calculation method may lead to overestimation of the value of team competitions; The investment return coefficient is divided according to the frequency and speed of effectiveness of the project, in order to evaluate the investment return rate of different sports projects. The comprehensive scoring model is constructed as follows:

$$S = p \times \max(0, C - x) \times n \times i \tag{14}$$

Among them, p represents the number of participants in the sport, x represents the number of winning cards in 2024 in the sport, C is a constant of 3, and n represents the number of events in the sport. i is the investment return coefficient. In the formula, if $max(0, C - x)$ is 0, it indicates that there will be a large number of medals won in 2024 and little room for improvement. If it is greater than zero, it indicates that there is a significant room for improvement.

The investment effectiveness coefficient i is divided into three levels. For sports with large teams, long investment cycles, or traditional challenges, such as basketball and volleyball, the value of i is 0.8; For most common exercises, the average effect is moderate, such as fencing, boxing, and i taking 1; For fast-paced or trendy sports such as surfing and breaking, the value of i is 1.2.

We calculated the comprehensive scores of various sports in France, Australia, and Germany respectively, and selected the sport with the highest scores for investing in a 'great coach'. The results are shown in the following table:

Table 13: Sport with the highest scores

Results	FRA	AUS	GER
Recommended items	Women	Women	Women
Sport	Football	Football	Hockey
2024 participants	20	19	18
2024 medals	0	0	0
2024 α	20.67	20.67	16.67
Comprehensive score	992.00	942.40	900.00

From the above table, it can be seen that the recommended investment sports for France

and Australia are women's football, while the recommended investment sports for Germany are women's hockey. Through observation, we found that these projects all have one thing in common: multiple participants, no medals won in 2024, and a large number of events set up, highlighting their great potential for winning awards.

Finally, we estimate the impact of investing in a 'great coach' on the average medal gain of these three countries in 2028. Taking the average medal gain calculated in 6.1 as a reference, take the average medal gain of different countries and sports as the mean:

$$\bar{\alpha} = \frac{1.000 + 1.375 + 1.071 + 1.698}{4} = 1.286 \tag{15}$$

The final average medal scores for France, Australia, and Germany can be calculated as:

$$\begin{cases} \alpha_{FRA} = 20.67 \times 1.28601 = 26.5818267 \\ \alpha_{AUS} = 20.67 \times 1.28601 = 26.5818267 \\ \alpha_{GER} = 16.67 \times 1.28601 = 21.4377867 \end{cases} \tag{16}$$

The average medal scores of the three countries in 2028 will all significantly be improved, reflecting the rationality and correctness of investment.

7 Other Original Insights

7.1 The Matthew Effect in Olympic Medal Allocation

By analyzing historical data from 1984 to 2024, we observed a significant and continuously intensifying Matthew effect in the distribution of Olympic medals. Specifically, the medal share of the top ten countries has continued to rise from 58% in 1984 to 73% in 2024, and is expected to reach 74% in 2028. This trend reflects the characteristic of "the strong always remain strong" in the development of competitive sports, as shown in the following figure:

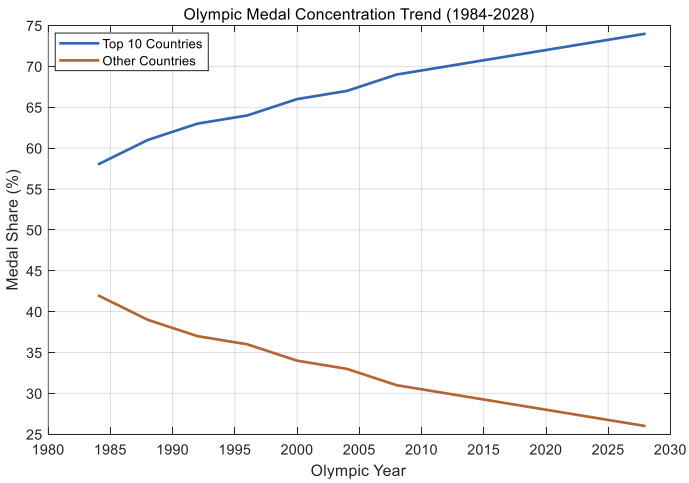


Figure 12: The Matthew effect in Olympic medal allocation

In depth analysis reveals that this effect mainly stems from three aspects: the cumulative effect of high-quality training resources, the maturity of the athlete training system, and differences in sports technology investment. For a sports powerhouse, our suggestion is to continuously increase investment in technological innovation, maintain competitive advantages, and

strengthen the training of young reserve talents to ensure the sustainability of advantages. For developing sports countries, it is recommended to adopt a "key breakthrough" strategy, concentrate resources on specific projects, introduce advanced training concepts, and improve the talent training system.

7.2 Time Dynamic Characteristics of the "Great Coach" Effect

In the process of studying the "Great Coach" effect, we found that there is a clear time-dependent characteristic of this effect. In the previous study, it was found that the average gain coefficient of the coaching effect is 1.28601, but this effect exhibits significant temporal dynamics:

- **Adaptation period** (0-1 year): The gain effect gradually appears, with an average increase of about 0.8 times.
- **Golden period** (2-3 years): The effect reaches its peak and can achieve a 1.5-2-fold improvement in performance.
- **Stable period** (4-5 years): The effect begins to stabilize and remains at around 1.2 times.
- **Decay period** (after 6 years): The effect gradually weakens and drops to below 1 times.

The temporal dynamic characteristics of this effect can be more clearly seen from the following figure:

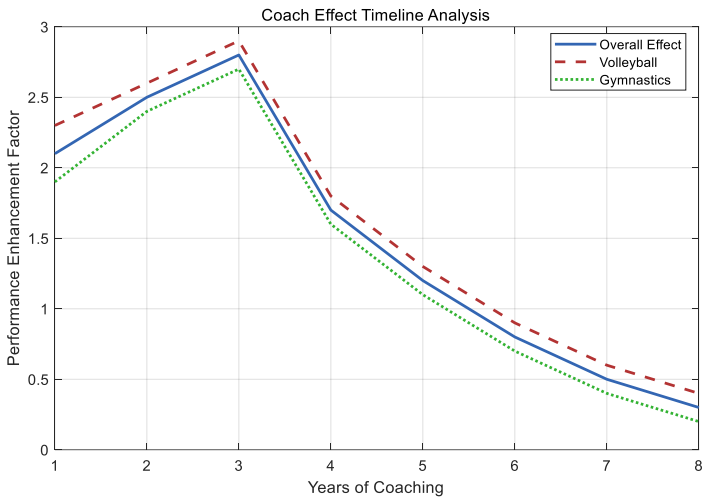


Figure 13: Coach effect timeline analysis of different sports

This pattern is common in different projects, but the intensity varies: team projects (such as volleyball) have a higher peak effect, but also a faster decay; The effect of individual events (such as gymnastics) is relatively gentle, but the duration is longer. For National Olympic Committees, **we recommend a signing period of 4-5 years for coaches**. To cover the entire Olympic cycle, the formation of the coaching team should be completed 2-3 years before the Olympics, and consideration should be given to setting performance evaluation standards based on time periods. At the same time, it is recommended to equip local assistant coaches to accelerate the transition period, establish a stable training system, and reduce the fluctuations caused by coach turnover.

8 Sensitivity Analysis

In this paper, we used a multi-layer perceptron (MLP) regression model based on CGO algorithm to predict the number of medals and gold medals in the Olympic Games. Due to the fact that the MLP model contains multiple input features, we performed separate random permute operations on each feature to evaluate their impact on the model's prediction results. We first calculated the original error of the model on the test set as the baseline error. Subsequently, we randomly shuffle each feature individually and recalculate the model's prediction error (MAE). The increment of error (compared to the baseline) can be considered as a measure of the importance of the feature to the model. In order to facilitate the observation of the incremental error size after scrambling various features, we have plotted the MAE images of the input features scrambled for medals and gold medals, as shown in the following figure:

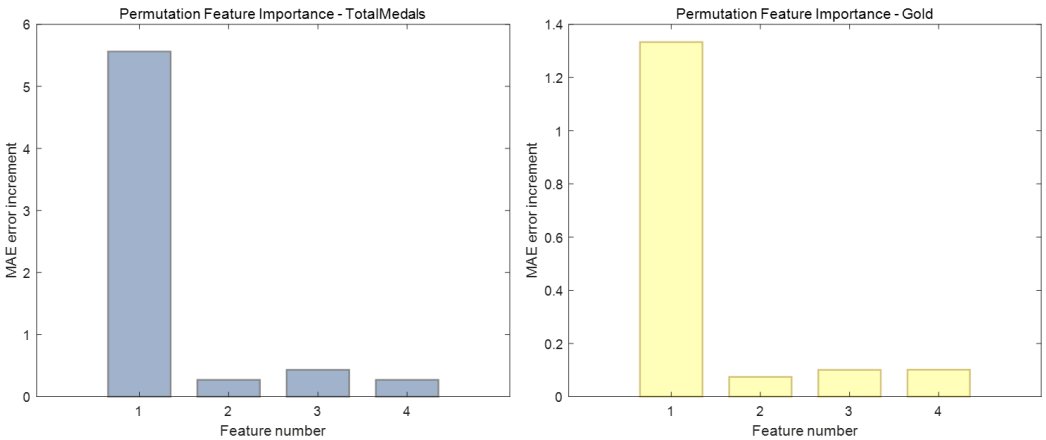


Figure 14: Permutation feature importance

The results show that both the total medals prediction model and the gold medals prediction model are extremely sensitive to the feature of ‘number of athletes’ and almost dominate the current feature set. Disrupting it can lead to a significant increase in prediction errors. It can be considered that the model prediction results are most sensitive to the number of athletes. If there is noise or statistical error in the "number of athletes" in the data itself, it is likely to significantly affect the model prediction. And the other three characteristics (especially whether they are the host) play a supportive or minor corrective role.

9 Model Evaluation and Further Discussion

9.1 Strengths

1. **Taking into account multiple factors:** The model comprehensively considers the impact of factors such as the number of athletes, continuous participation experience, host advantage, and total number of events on medal count, which can more comprehensively reflect the performance of each country in the Olympic Games.
2. **Using advanced algorithms:** MLP regression model and CGO chaotic game optimization algorithm are introduced, which can capture nonlinear relationships and complex interactions in the data, improving the accuracy and reliability of predictions.

3. **Providing confidence intervals:** The model not only predicts the number of medals, but also provides confidence intervals at different confidence levels, providing a quantitative evaluation of the uncertainty of the prediction results and enhancing the practicality and credibility of the model.
4. **Dynamic programming solution:** When dealing with Poisson binomial distribution problems, dynamic programming methods are used to effectively avoid the complexity of direct calculations and improve computational efficiency.

9.2 Weaknesses

1. **Data limitations:** Despite the use of multiple factors in the model, the limitations of the data still exist. For example, certain countries may have unique advantages or disadvantages in specific projects, which may not have been fully considered.
2. **Model complexity:** MLP regression models and CGO optimization algorithms have high complexity and require a large amount of computational resources and time for training and optimization, which may not be suitable for real-time prediction or resource limited scenarios.

9.3 Further Discussion

1. **Model fusion:** It is possible to consider fusing multiple models (such as linear regression, MLP regression, random forest, etc.) to improve the accuracy and stability of predictions through ensemble learning methods.
2. **Introducing more features:** It is possible to consider introducing more factors that affect the number of medals, such as the country's sports culture, training facilities, sports policies, national economic situation, etc., to further improve the predictive ability of the model.

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